Experiment-9

To Implement Travelling Salesman Problem

Aim

To develop a Python program to solve the Travelling Salesman Problem (TSP) using the Brute Force approach.

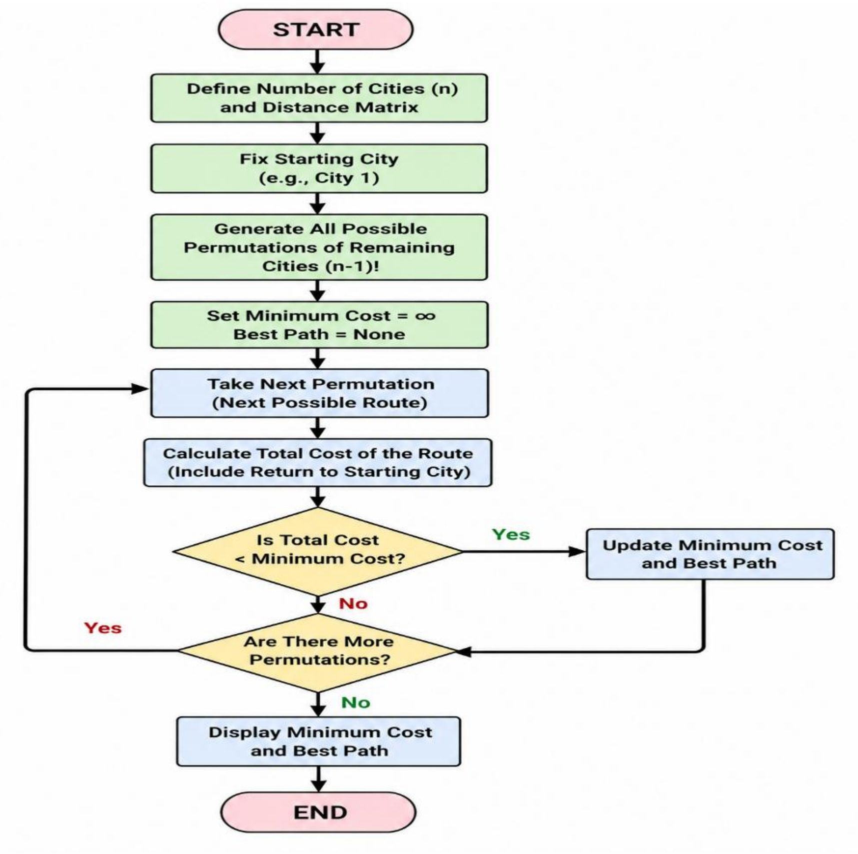
Objective

- To implement the Travelling Salesman Problem (TSP).
- To find the minimum cost path that visits every city exactly once and returns to the starting city.
- To understand optimization techniques in Artificial Intelligence.

Algorithm

1. Start the program.
2. Define the distance matrix for all cities.
3. Generate all possible routes using permutations.
4. Calculate the total cost of each route.
5. Compare the route cost with the minimum cost.
6. Update the minimum cost if a shorter route is found.
7. Display the minimum travel cost.
8. Stop the program.

Flowchart:

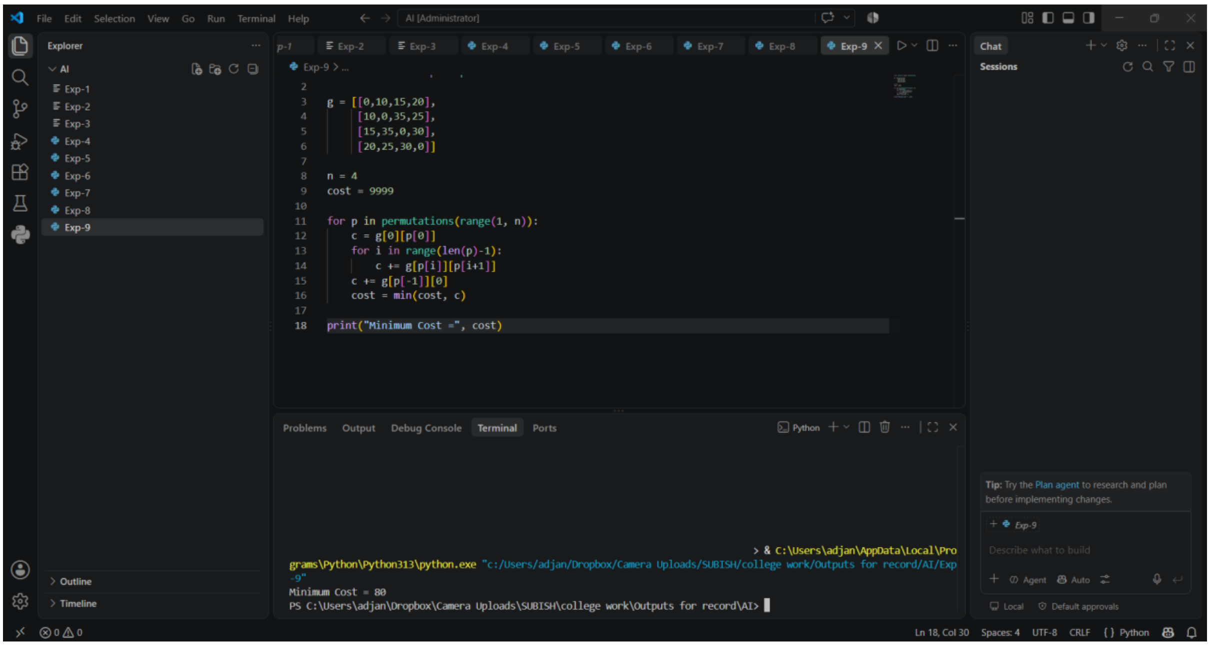


Code:

```
from itertools import permutations


g = [[0,10,15,20],
     [10,0,35,25],
     [15,35,0,30],
     [20,25,30,0]]


n = 4
cost = 9999


for p in permutations(range(1, n)):
    c = g[0][p[0]]
    for i in range(len(p)-1):
        c += g[p[i]][p[i+1]]
```

```
    c += g[p[-1]][0]
    cost = min(cost, c)

print("Minimum Cost =", cost)
```

Output:



Result:

The Travelling Salesman Problem (TSP) was successfully solved, and the minimum travel cost was obtained.

Conclusion:

The Travelling Salesman Problem was solved by evaluating all possible routes and selecting the one with the minimum travel cost.